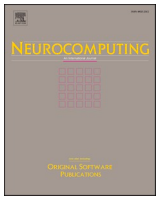
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Granular transfer learning

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ABSTRACT

Transfer learning is aimed at supporting the design of machine learning models in the target domain D_t, given that the knowledge (model) has already been constructed in the source domain D_s. The domains D_t and D_s (as well as the corresponding tasks T_s and T_t) are similar, yet not identical. As a result, the model transferred from D_s to D_t in this new environment exhibits its relevance (credibility) only to some limited extent. In this study, we develop an original approach, where we advocate that the knowledge transfer (model transfer) gives rise to a granular model where the level of information granularity associated with the produced results quantifies the relevance (quality or credibility) of the transferred model. In other words, we stress that the quality of knowledge transferred to D_t becomes captured through a granular generalization of the original numeric model. The overall systematic design process is elaborated on by focusing on the development process of granular neural networks carried out on a basis of the numeric neural networks coming from D_s. The key aspect of the design is to elevate the existing numeric neural network to its granular counterpart by admitting that the connections of the developed model come in the form of information granules, in particular intervals and fuzzy sets. The optimization process is guided by adjusting (optimizing) the level of information granularity being regarded as an essential design asset. The optimized performance index builds upon the descriptors of information granules commonly encountered in Granular Computing. In particular, coverage and specificity measures are treated as sound performance indicators of the quality of knowledge transfer (viz. the performance of the granular neural network expressed in the target domain). Several illustrative examples are provided to visualize the performance of the established design environment.

1. Introduction

Transfer learning has been an intensively studied paradigm of machine learning [1,2]. The essence of this learning strategy is to extract previously acquired knowledge (model) and apply it to a new environment. The commonly used terminology concerns here a source domain D_s in which the knowledge has been acquired and target domain D_t in which the already acquired knowledge (model) becomes used. Along with the domains associated are tasks, T_s, and T_t, respectively. They could be of any nature, in particular classification or prediction. Thus the structures involved in transfer learning are pairs (D_s, T_s) and (D_t, T_t).

There are a number of motivating factors behind the use of transfer learning, especially

(i) the small data set available in the target domain (which does not support designing advanced models equipped with a large number of parameters), (ii) enhancements of the robustness of the ML model, and (iii) elimination of cold start problem [1].

Transfer learning is also referred to as knowledge reuse, learning by analogy, domain adaptation, and pretraining, among others. Conceptually, this type of learning is illustrated in Fig. 1(a).

Depending on the nature of data in D_t, there are two main practical scenarios behind the involvement of transfer learning in the design of neural networks:

(i) Large data sets of high dimensionalities. Similar in their characteristics to data in the source domain, these data are used in the

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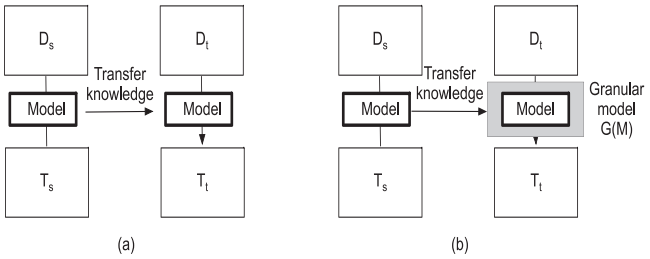


Fig. 1. Transfer learning (a) and granular transfer learning: a paradigm shift (b) Note the emergence of a granular model in the target domain.

target domain. Transfer learning helps reduce computing overhead as the model is considered as is, and no learning is involved. Obviously, the larger the differences between the target and source domains, the more visible the deterioration of the quality of the model becomes present. Consequently, this situation calls for an efficient way of quantifying the usefulness (credibility) of the produced results. This scenario is in line with the realization of the paradigm of green AI (green machine learning), where computing efficiency (sustainability) is strongly advocated, along with a substantial emphasis placed on the efficiency of the design of machine learning architectures and their properties of interpretability and brittleness.

(ii) Small data set in D_t, manifesting especially in terms of the number of data points. In this situation, the design of models, in particular, those with a large number of parameters (namely, connections of neural networks), is difficult, if not infeasible. Likewise, for small data associated with high dimensional space, the design of even linear regression models could be questionable (e.g., because of the computational stability of matrix manipulation), and the credibility of such model must be carefully quantified. Small data are encountered in practice in cost estimation situations when dealing with new software projects: very few data are acquired, and one has to rely heavily on the models formed for previously completed similar projects. We have to become aware that the less similar those projects are to the one considered now, the more caution has to be exercised when relying on the results produced by the transferred model.

There are two modes of the development of models (and neural networks as quite commonly encountered architectures) in the target domain:

(i) *passive approach*. The model constructed in the source domain is transferred to the target domain without any additional learning. However, considering the differences between the target and source domains, the model has to be endowed with some measure of credibility of results generated by it in the target domain. With the decreasing resemblance (closeness) between the two domains, such quantification becomes paramount as it sets up some tangible limits within which the transferred model exhibits some usefulness.

(ii) *active approach*. Here, a variety of design strategies are employed to facilitate a new model, in which the design is actively supported by the model transferred from the source domain in the form of some pre-training. To reduce a substantial computing overhead, the design of smaller models is considered; the strategy is explored under the banner of knowledge distillation, forming one of the variants of transfer learning.

Deep transfer learning is a way of learning a model (typically deep neural networks) that leverages knowledge from other domains [3]. In recent years, deep learning techniques in neural networks have been successfully applied to a wide range of applications, including image processing [4–8] and time series analysis [9–13], among others. Given the significant computing overhead associated with the learning methods, there is a growing interest to reduce the amount of optimization time.

It is evident that once the source and target domains become more distinct, the knowledge (model) transferred from D_s to D_t starts exhibiting lower relevance and usefulness. To quantify this phenomenon, we advocate that the numeric model (say, a neural network) transferred to D_t has to be endowed with the mechanism capturing and quantifying the effect of such diminished usefulness. In this study, we pursue a novel direction of investigation in transfer learning by postulating that a *numeric* neural network constructed in D_s becomes a *granular* neural network when applied in D_t. We exploit the principles of Granular Computing [14,15] to develop a granular model as a sound augmentation of the transferred knowledge. This is done in two main ways. First, we show that the differences between D_s and D_t are reflected by elevating the type of information granularity by proceeding from the type-0 information granule (numeric result) (present in Ds) to the type-1 information granule present in D_t. For the sake of completeness, let us recall that by information granules of type-0, we mean numeric entities (and numeric models), whereas type-1 information granules are those formalized as intervals, sets, fuzzy sets, and probabilities, see [14–17]. Second, we design a granular model in D_s by optimizing an objective function capturing the numeric data vis-à-vis information granules emerging through the granular construct in D_s. It is worth noting that while fuzzy sets were studied in the setting of transfer learning, see [18, 19], the approach advocated here is novel and has not been investigated so far.

Numerous formalisms describe and process information granules. In this study, the term information granule is sought in a general manner (realized as intervals, fuzzy sets, rough sets, etc.) when considering the underlying concepts. However, to enhance the clarity of the presentation, we concentrated on interval information granules when elaborating on algorithmic developments. The key contributions of this study include:

- Introduces a new approach to transfer learning with Granular Computing, creating networks that evaluate cross-domain knowledge credibility.
- Proposes a novel framework to quantify transferred knowledge quality through information granularity, enhancing transfer learning measurement.
- Leverage granular computing to optimize target domain neural network performance, using adjustments in information granularity for improved adaptability.

The motivation for using a granular model is driven by the need to effectively manage imprecision, enhance interpretability, provide flexibility and adaptability, and improve knowledge representation. In particular, the quality (credibility) of any result produced by the granular model can be quantified when the original model has been transferred to the target domain.

In this study, we propose an original granular extension of transfer learning by showing how the neural network built in D_s is transferred with the level of information granularity aimed at its usage in D_t. The essence of the approach is highlighted in Fig. 1(b). We relate the level of information granularity with the "distance" (resemblance) between the two domains. This gives rise to the augmentation of the model by its granular wrapper (shaded box in Fig. 1b)). Considering the above general taxonomy, the developed study presented here focuses on the passive approach and a small data set present in the target domain.

The study is structured as follows. We start with a brief overview of transfer learning (Section 2). This is followed by constructing granular neural networks (Section 3). Next, we elaborate on the role of information granularity and its optimization in developing the originality of the granular model built-in D_s and then transferred to D_t. Detailed

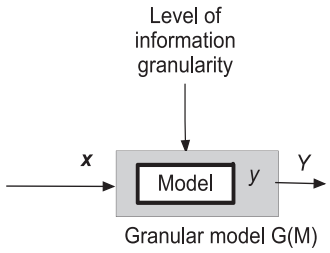


Fig. 2. From numeric to granular model: optimized level of information granularity ε serves as a design asset.

computing completed with interval information granules is covered in Section 4. Experimental studies are presented in Section 5, while conclusions are covered in Section 6.

2. Transfer learning

Transfer learning carried out in neural networks and deep learning is a methodology and a slew of algorithms that allow a model trained on one task or domain to be applied to a different task or domain. This can be especially motivated when there is a lack of data or resources available for training a model on the target domain. Transfer learning can be classified into several categories:

1. Network-based transfer learning: In this approach, the network is trained on a source domain as part of a pre-training process. The network's parameters are then fine-tuned on the target domain. Pre-trained networks, such as ImageNet [20] and AlexNet [21], have been used as the initial weights for many vision applications [22,23] due to the limited size of the target domain. More sophisticated models like ResNet [24], and transformers [25] have significantly advanced the field. These models have been pre-trained on vast datasets, capturing a wide range of features that can be effectively transferred to various tasks, including those in natural language processing and computer vision.
2. Adversarial-based transfer learning: This approach uses generative adversarial networks [26] to find portable compact representations that are appropriate for both the source and target domains. Features from both domains are sent to an adversarial layer for domain adaptation in a feed-forward neural network [27]. There is ongoing research into reducing domain discrepancy more effectively, ensuring that the features generated by the adversarial process are not only indistinguishable but also relevant and useful for the target task [28].
3. Instance-based transfer learning: In this approach, weighted instances from the source domain are selected as additional instances in the target domain [3]. TrAdapt [29] is a popular method of transfer learning where instances of the source domain are weighted according to their similarity to the target domain. This method has been applied to image processing [30]and time series analysis [31]. There is a growing interest in combining instance-based methods with other forms of transfer learning to leverage both fine-tuned pre-trained models and selected instances for improved performance on the target task [32,33].
4. Mapping-based deep transfer learning: In this approach, samples from the source and target domains are mapped into a new data space [3]. Efforts have been made to create more sophisticated shared spaces that better capture the underlying structure of both domains, improving the model's ability to generalize from source to target [34].

Network-based transfer learning, which uses pre-trained models like ImageNet, excels in performance and efficiency for many tasks, but it can be demanding in terms of resources and may not always adapt well to different domains. Adversarial-based transfer learning, using GANs, effectively bridges the gap between domains, making features more relevant for the target task, though it can be computationally intense and unstable. Instance-based transfer learning is straightforward and adaptable, making the most of available data by selecting relevant instances, but its success hinges on the similarity between the source and target domains and can be costly in terms of computation. Lastly, mapping-based deep transfer learning enhances generalization by creating new data spaces, offering better representation and flexibility, yet it involves complex implementation and risks overfitting if there isn't enough data.

The focus on making transfer learning more efficient, effective, and applicable across a broader range of domains and tasks underscores the rapid evolution and growing importance of this area within the broader field of machine learning and Artificial Intelligence.

3. Construction of granular model

3.1. A. The underlying idea: a granular wrapper

Once a numeric model developed for the source domain has been constructed, transferring it to the target domain becomes advantageous because of a number of arguments outlined above. However, one has to become aware that D_s and D_t are different, and while the "distance" (lack of similarity) between these domains becomes larger, it becomes indispensable to quantify the performance of the transfer (model functioning in D_t) and the quality of the transferred constructs. This is realized by admitting that model M (built-in D_s) becomes granular. The granular model quantifies the performance of such a model in D_t. The granular transferred model can be symbolically denoted as $G(M)$ to underline that this construct is endowed with a granular wrapper. In other words, we introduce the following transformation.

$$\mathrm{M} \xrightarrow{\varepsilon} \mathrm{G(M)} \tag{1}$$

where ε stands for a non-negative level of information granularity whose value reflects the closeness (resemblance) between D_t and D_s. Refer to Fig. 2, which highlights the key point of the introduced methodology.

The level of information granularity can be considered an essential design asset facilitating the realization of the granular model. From this point of view, one may regard the granular model as a wrapper of the numeric model, elevating its abstraction level by accommodating the level of information granularity. The granular model built on the basis of M gives rise to granular outputs and, in this way, quantifies the performance of the model now functioning in the target domain.

The optimization process aims to determine the level of information granularity so that a certain performance index becomes optimized (the detailed optimization approach is discussed in Section 3).

3.2. B. Information granularity in the transferred models

In light of the transformation (1), for any $\boldsymbol{x}$ in D_t the granular model $G(M)$ produces information granule Y

$$Y = G(M)(\boldsymbol{x}) \tag{2}$$

Considering that M is endowed with some numeric parameters $\boldsymbol{a}$ (whose values have already been optimized on D_s), namely $y = \mathrm{M}(\boldsymbol{x}; \boldsymbol{a})$, $\dim(\boldsymbol{a}) = p$, the following two classes of transformations producing granular constructs $G(M)$ are contemplated.

(i) *Granular parameters*. The original vector of parameters $\boldsymbol{a}$ is elevated to its granular counterpart by admitting some level of information granularity ε as described symbolically in the following way: $\boldsymbol{a} \xrightarrow{\varepsilon} \boldsymbol{A} = G(\boldsymbol{a})$. The coordinates of $\boldsymbol{A}$ are regarded as information granules spanned around the numeric values of the corresponding parameters. This apparently makes the model

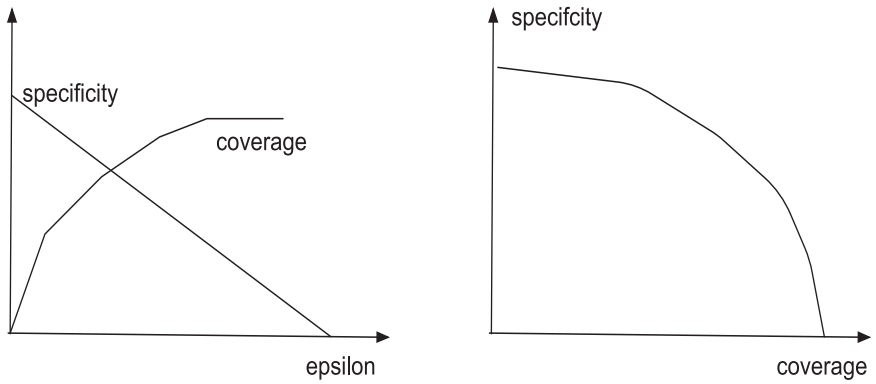


Fig. 3. An example plot of coverage and specificity treated as functions of ε. The optimal value ε is attained when the product $\overline{\text{cov}} * \overline{sp}$ attains maximal value.

granular, and the result of the model is an information granule itself.

(ii) *Granular results* of the model. Keeping the original numeric values of the parameters, the numeric output y is elevated to its granular version Y following the transformation

$$y = M(\boldsymbol{x}; \boldsymbol{a}) \xrightarrow{\varepsilon} Y = G(\mathrm{M}(\boldsymbol{x}; \boldsymbol{a})) \tag{3}$$

where, again, a certain level of information granularity ε is involved.

Proceeding with (i), the transformations applied to a single parameter a gives rise to the corresponding interval A in the form

$$A = [\min(a(1\text{-}\varepsilon), a(1+\varepsilon)), \max((a(1\text{-}\varepsilon), a(1+\varepsilon))], \varepsilon \in [0,1] \tag{4}$$

$$A = [\min(a/(1+\varepsilon), a(1+\varepsilon)), \max(a/(1+\varepsilon), a(1+\varepsilon))], \varepsilon \geq 0 \tag{5}$$

The level of information granularity is involved in the formation of information granules. How to proceed from their numeric to granular augmentations is realized by invoking protocols of allocation of information granularity.

The optimal value of ε is determined by the optimization of the performance index (objective function), which involves two components of the obtained information granule such as coverage and specificity. Let us consider D_s consisting of pairs $(\boldsymbol{x}_k, target_k)$, k=1, 2,…,N

Coverage The underlying idea is to produce such granular output Y so that for some $\boldsymbol{x}$ it includes the corresponding target. This requirement is quantified by the coverage criterion using which one counts the number of cases $target_k$ is included in the corresponding Y_k, Y_k=$G(M)(\boldsymbol{x}_k)$. Formally, coverage, *cov*, is computed as

$$cov = incl(target_k, Y_k) \tag{6}$$

where the inclusion predicate, *incl*, returns 1 if $target_k$ is included in $Y_{\boldsymbol{k}}$ otherwise, the returned result is zero.

Specificity We also require that Y is highly specific, meaning that its precision is high. The specificity *sp* is regarded as a decreasing function of the "length" of the information granule. In the case of an interval [*a, b*], it is a decreasing function of the difference *b-a*. In particular, one could express *sp* as a linearly decreasing function of this difference (length) in the form

$$sp([a, b]) = \max(0, 1\text{-}(b\text{-}a)/range) \tag{7}$$

where *range* is the range of values assumed by the output and serves as a certain calibration factor.

The higher the value the *cov*, the better the granular model in terms of its ability to "cover" the numeric data. Obviously, the higher the value of ε, the broader the resulting interval and higher likelihood of satisfying the coverage requirement. Higher values of ε yield lower specificity *sp* of the results. Following the principle of justifiable granularity [35,36], we maximize the product of coverage and specificity by determining an optimal value of ε.

The performance of the granular model can be studied by analyzing the values of the average coverage and specificity for various values of the level of information granularity ε. In other words, we consider

$$\overline{\text{cov}} = \frac{1}{N}\sum_k \text{cov}(Y_k) \overline{sp} = \frac{1}{N}\sum_k sp(Y_k) \tag{8}$$

where N = card (D_t). Some illustrative plots of such relationships are presented in Fig. 3. In general, as noted earlier, increasing values of ε result in higher coverage and lower values of specificity.

The third approach (protocol 3) is to make the output granular by building an interval around the result produced by the numeric model. For $\boldsymbol{x}_k$ and $target_k$ in D_t, we calculate an error produced by the model M constructed in D_s

$$\delta_k = M(\boldsymbol{x}_k) - target_k \tag{9}$$

and approximate the error as a polynomial P of $|\delta_k|$, say $P(\delta_k)$. The interval output of protocol 3 for any $\boldsymbol{x}$ in D_t is expressed as

$$\boldsymbol{Y} = [M(\boldsymbol{x}) - \varepsilon\, P(\delta), M(\boldsymbol{x}) + \varepsilon\, P(\delta)] \tag{10}$$

where the level of information granularity ε is optimized in the same way as before, viz., our objective is to maximize the production of average coverage and specificity. The first two protocols ((4) and (5)) are concerned with the parameters of the models, while the third one (10) focuses on the generation of granular output built on the basis of the numeric output.

Apparently, the two requirements of coverage and specificity are in conflict; see also Fig. 3.

For instance, in case (7), the maximal specificity value is obtained for ε=0, while this value produces the zero coverage value. Thus, the product of these two requirements is a sound compromise.

The optimization (maximization) of the $\overline{\text{cov}}$ is, and sp can be sought as a two-objective optimization carried out with respect to e, or we can transform it to a scalar optimization problem by maximizing the product of coverage and specificity, viewing it as a function of ε. This leads to the problem $\text{Max}_\varepsilon V$, where

$$V(\varepsilon) = \overline{cov} * \overline{\text{sp}} \quad \text{and} \quad \varepsilon_{opt} = \arg\ \max_\varepsilon V(\varepsilon) \tag{11}$$

By following the plot in Fig. 3, an optimal value of ε is determined. Also, the global performance of the granular model can be determined by taking the area under the curve, as illustrated in Fig. 3. For (11), one computes

$$V_{\text{total}} = \int_0^1 (\overline{cov} * \overline{\text{sp}}) d\varepsilon \tag{12}$$

Having the three allocation protocols of information granularity, one can quantify the result of transfer knowledge by determining the values of $V(\varepsilon_{\text{opt}})$ and V_{total}. These indicators reflect the quality of the knowledge transferred from D_s to D_t. The more "remote" (D_s,T_s) and (D_t,T_t) are, the

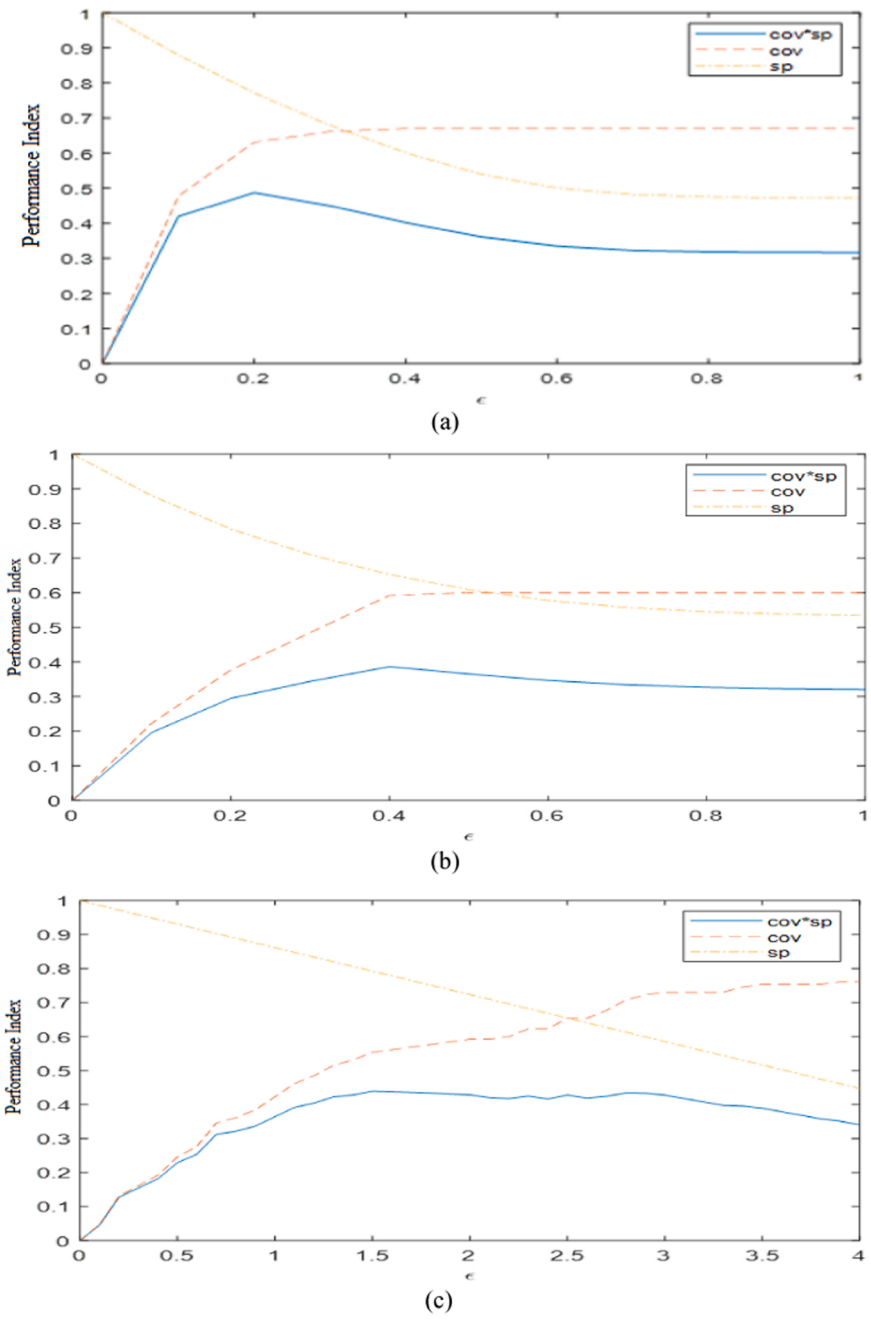


Fig. 4. Values of performance index V obtained for successive values of ε. Using (a) protocol 1, (b) protocol 2, and (c) protocol 3.

Table 1
maximum values of performance indices using A uniform distribution.

Protocol	Coverage	Specificity	V	ε_{opt}	V_{total}
1	0.63	0.77	0.49	0.2	0.30
2	0.63	0.79	0.50	0.4	0.31
3*	0.55	0.79	0.44	1.5	0.38

* $P(\widehat{y}_k) = 0.84\widehat{y}^3 + 50.17\widehat{y}^2 + 996.151\widehat{y} + 6586.7$ (see Fig. 4)

lower the values of the performance of the transferred knowledge (model). Along this line of thought, one could consider a criterion of admitting transfer knowledge as a suitable way to proceed:

We complete transfer knowledge from D_t to D_s if V, taken as the product of specificity and coverage (8) exceeds a certain threshold level $\tau\square$

The above process of building a granular model could be refined. Instead of a single value of the information granularity ε, each parameter of the model is impacted differently by coming up with its corresponding level of information granularity. Thus the optimization deals with the optimization of the p-dimensional vector of levels of information granularity $\boldsymbol{\varepsilon} = [\varepsilon_1\ \varepsilon_2\ \ldots\ \varepsilon_p]$, $\varepsilon_j \geq 0$ so that $V(\boldsymbol{\varepsilon})$ attains maximum. This optimization requires using population-based optimization methods, such as particle swarm optimization (PSO), genetic algorithm (GA).

4. Computing with information granules in neural networks

The objective of this part is to provide detailed computing within information granules (granular parameters) carried out for neural networks. It offers some helpful prerequisites as neural networks are experimented with in the experimental studies. As discussed, the pa-

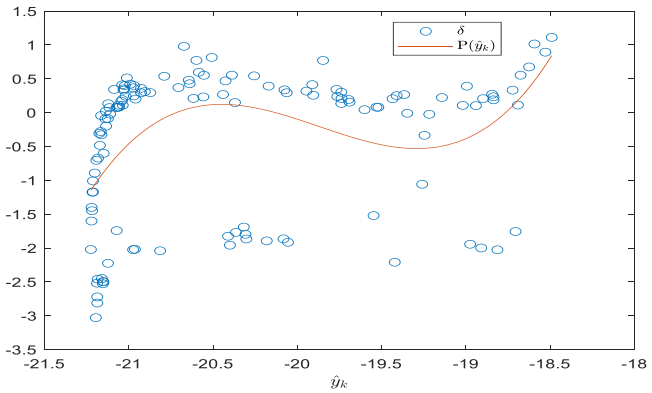


Fig. 5. Polynomial $P(\widehat{y}_k)$ of third degree.

rameters of the granular model are information granules. In this section, we elaborate on the detailed computing in case of neutral networks equipped with granular either interval of fuzzy set connections. We show a way in which information granularity is propagated through the layers of the networks. Consider a single output neural network with a single hidden layer composed of h neurons. The corresponding calculations were carried out as follows by proceeding with calculations starting from the hidden layer

$$z_i = f\left(\sum_{j=1}^{n} w_{ij}x_j + w_{i0}\right), i = 1, 2, \ldots, h$$
$$y = f\left(\sum_{i=1}^{h} u_i z_i + u_0\right) \quad (13)$$

The neurons are endowed with the weights (connections) w_{ij}, u_i and biases w_{i0}, u_0. f is a nonlinear function (typically a sigmoidal or tanh one). If we admit interval-valued weights W_{ij} and U_i, the calculations produce an interval Z_i and then an output interval Y. Computing with interval information granules becomes advantageous from a numerical point of view [36–38]; we directly follow the calculus of interval mathematics with the well-known and computationally straightforward calculations carried out on intervals, say

addition $A + B = [a+c, b+d]$

multiplication $A*B$ = [min (ac, ad, bc, bd), max (ac, ad, bc, bd)]

subtraction A-$B = [a-d, b-c]$

division $A/B = [a, b] * [1/d, 1/c]$ (assuming 0 is not included in $[c,d]$)

where $A = [a, b]$ and $B = [c, d]$ are two numeric intervals.

Likewise, given a monotonically increasing function f (say, a sigmoidal function) the result of transforming $[a, b]$ via f results in the interval $[f(a), f(b)]$. For the monotonically decreasing function, one has $[f(b), f(a)]$.

Computing with information granules represented in the form of fuzzy sets becomes more computationally demanding given the fact that the form of membership functions is usually not retained under various arithmetic operations, except for addition and subtraction. For instance, adding two triangular fuzzy numbers yields a triangular fuzzy number, but the result of the multiplication of two triangular fuzzy numbers is no longer a triangular one. In the case of probabilistic information granules, in successive computations, one follows the calculus of random variables [39]. The overall design process can be summarized as:

Given: data Ds:{(x_k, $target_k$) | k=1,2,…, N_s}; D_t = {(x_k, $target_k$) | k=1,2,…, N_t}

Step 1: Design the model M on the source domain D_s

$\boldsymbol{y}$ = M($\boldsymbol{x}$; $\boldsymbol{a}$)

train_model_on_source_domain(Ds)

a = get_model_parameters(M)

Step 2: for each level of information granularity ε:

Construct interval-valued parameters (a) following protocols in (4), (5), and (10)

interval_a = construct_interval_valued_parameters(a, ε)

Step 3: Transfer the model M to the target domain Dt and obtain the interval output Y

interval_M = transfer_model_to_target_domain(M, interval_a)

interval_Y = interval_M(Dt)

Step 4: Compute the performance indices cov, sp, and V

cov = compute_coverage(interval_Y, Dt)

sp = compute_specificity(interval_Y, Dt)

V = compute_performance_index(cov, sp)

Quantify the protocol's performance by determining $V(\varepsilon_{opt})$ and $Vtotal$ for both uniform and non-uniform levels of information granularity

Step 5: Output the results output $V(\varepsilon_{opt})$ and $Vtotal$

5. Experimental studies

The reported series of experiments completed on publicly available data quantifies the performance of granular transfer learning. The experiments are set up in different scenarios by studying the impact of the various aspects of the environment on the performance of the target environment. The original data are split into the source and target parts (domains) D_s and D_t, where the size of data in D_t is far smaller than the size of data in D_s, say around 5 %. The model is constructed and

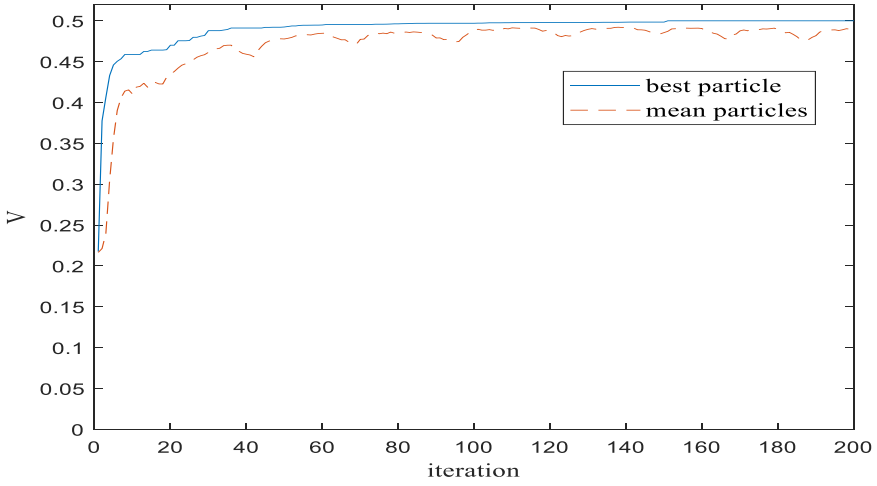


Fig. 6. Convergence of PSO in successive iterations.

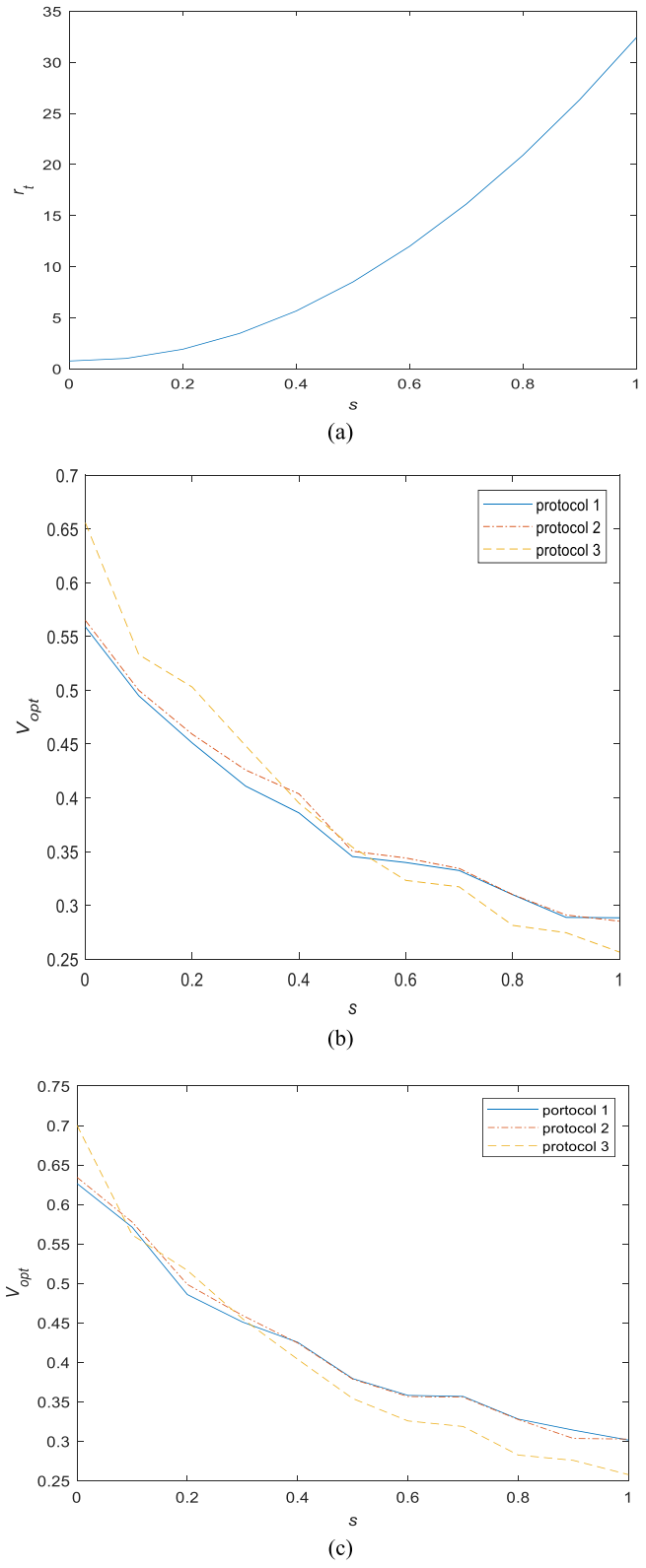


Fig. 7. Performance index as a function of σ for (a) ratio r_t (b) V_{opt} (uniform ε) and (c) V_{opt} (non-uniform ε) *s*:x-axis represents the variation of σ from 0 to 1.

evaluated in the D_s, where the training and testing data parts are split in the proportion of 70–30 %. Then, the model is transferred to the target domain, where its granular generalization is formed using the protocols described in Section III. The performance index is taken as a mean squared error (MSE) Q computed over the corresponding training or

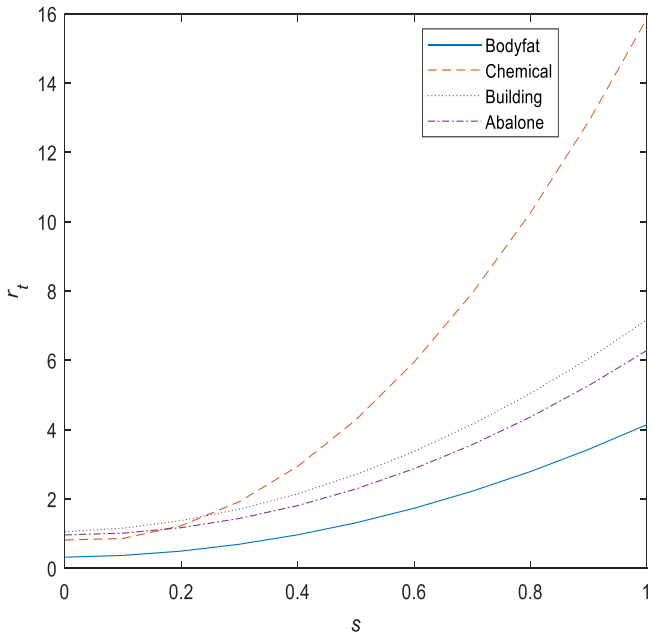


Fig. 8. Plot of r_t regarded as a function of σ for datasets Bodyfat (252×15, h=8), Chemical (498×8, h=4), Building (4208×14, h=10), Abalone (4177×8, h=4).

testing data, namely

$$Q = \frac{1}{N}\sum_{k=1}^{N}(y_k - t_k)^2 \tag{14}$$

The output variable (t_k) is normalized. The models constructed on D_s is taken as a multilayer neural network with only one hidden layer whose values of the main structural parameters (viz. the number of layers and the number of neurons in each layer) are determined experimentally. Along with the granular model transferred to D_t, using D_t, the model is constructed from scratch (again, for this purpose, D_t is split into a training and testing part). The values of the performance index Q on $D_{s,\text{train}}$, $D_{s,\text{test}}$ $Q(D_{s,\text{train}})$, $Q(D_{s,\text{test}})$ and $Q(D_{t,\text{train}})$ as well as $Q(D_{t,\text{test}})$, considered together serve as an indicator of the quality of the models constructed in D_t. More specifically, a concise description of the performance is obtained by comparing the ratios $r_s = Q(D_{s,\text{test}})/Q(D_{s,\text{train}})$ versus $r_t = Q(D_{t,\text{test}})/Q(D_{s,\text{train}})$. High values of r_t observed with respect to low values of r_s point at the weak performance of the models constructed directly on D_t. Along with the original data forming D_t, several of its modifications are considered by impacted them by some additive normal noise $N(0,\sigma)$ with several values of variance.

Example 1. We use the vinyl bromide data (16 inputs and 1 target) (https://data.world/datasets/vinyl). In the source environment D_s, the split of training and testing data yields 20,000 and 3000 data, while D_t consists of 100 and 30 data in the training and testing part with σ=0⊡3 characterizing the additive noise.

A neural network with a single hidden layer containing 10 neurons resulted in Q values of 0.40 and 0.41 on the training and testing on D_s, respectively. Increasing the size of the neural network did not improve performance. Therefore, all subsequent results were reported using a hidden layer of 10 neurons, which resulted in r_s and D_t values of 1.0 and 1.35, respectively. However, if the neural network were based on D_t, the performance would be much worse, with Q values of 0.24 for training, 2.23 for testing, and a high r_t value of 5.57.

The model directly learned on D_t exhibits poor performance when designed from scratch, as indicated by the corresponding value of rt (>>1). Although the ratio *rs* indicated the good performance of the model on Ds compared to rt, the performance of both models resulted in poor performance on D_t.

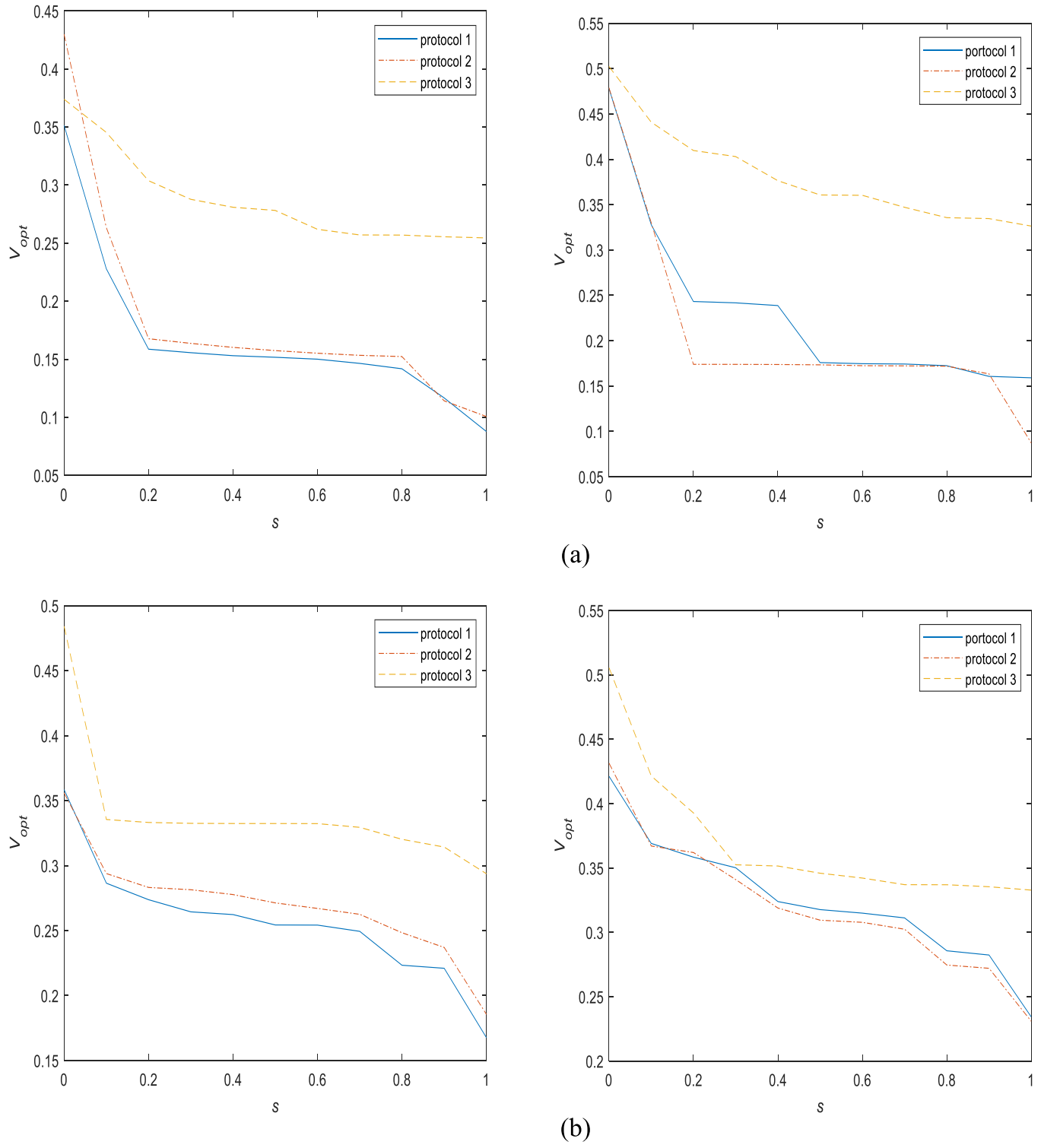


Fig. 9. The performance index *V*opt of uniform ε (left) and non-uniform ε (right) of public data sets (a) Bodyfat, (b) Chemical, (c) Building, and (d) Abalone.

The values of the average product of the coverage and specificity regarded as a function of e are illustrated in Fig. 4. Invoking the protocols of the design of the granular augmentations of the network gave rise to the best performance of uniform allocation of granularity, which is reported in Table 1. It should be noted that the approximation of error δ_k (9) is achieved through a polynomial of degree *n* that is the best fit (in a least-squares sense), where n is chosen to be 3 for simplicity, as shown in Fig. 5.

For D_t, the values of V_{total} are 0.30 (1st protocol), 0.31(2nd protocol), and 0.38 (3rd protocol), which resulted in the best global performance. The optimal value of ε for the individual protocols using the non-uniform level of information granularity with respect to the values of *V*: 0.53 (1st protocol), 0.53 (2nd protocol), and 0.45 (3rd protocol). The optimization of the levels of information granularity for each weight of the network is completed using particle swarm optimization PSO with the swarm size consisting of 200 particles, and a maximal number of

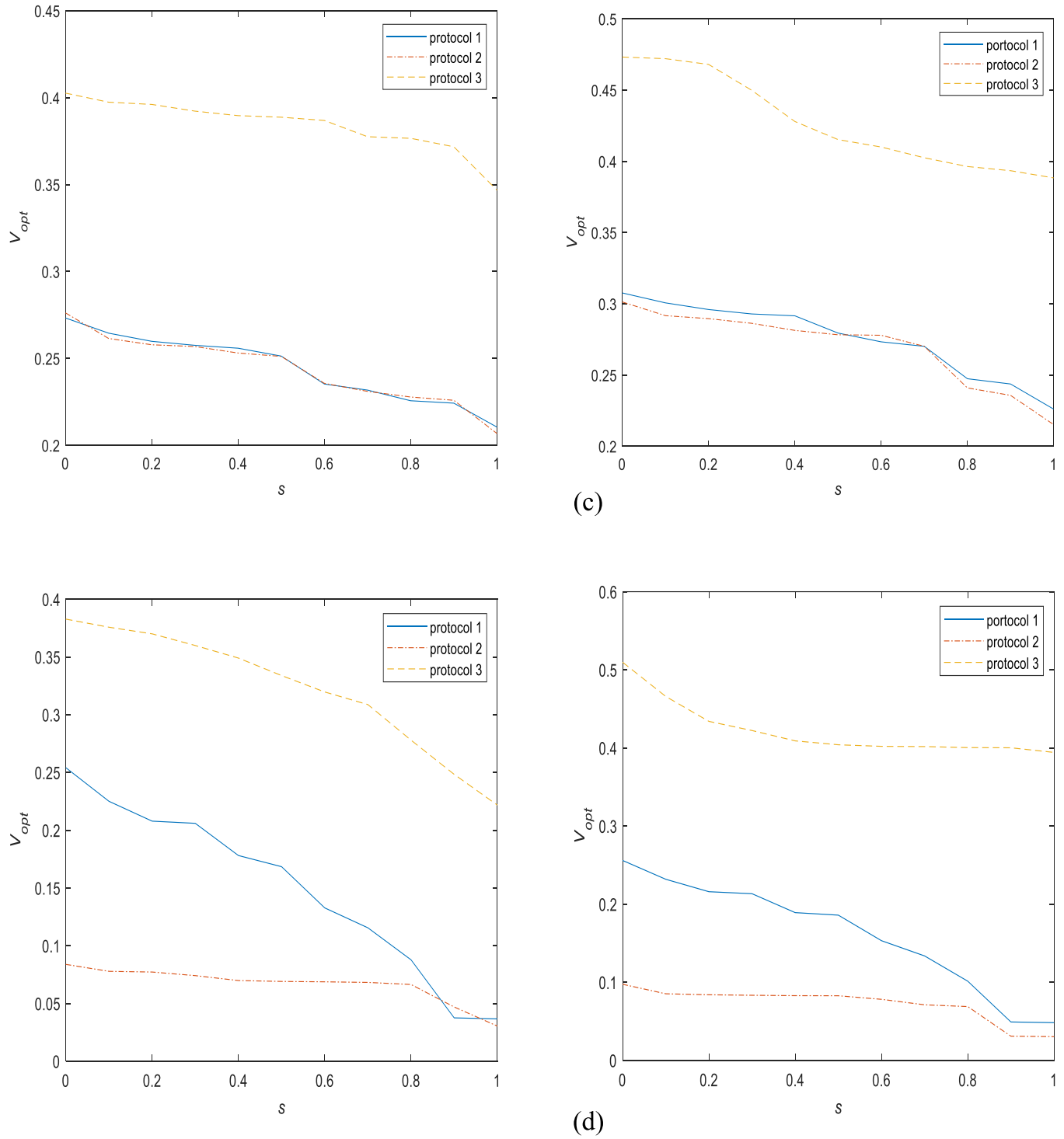


Fig. 9. (*continued*).

iterations is set to 200. In PSO, each particle represents a potential solution to the optimization problem. The algorithm continuously updates the position of each particle based on the positions of other particles in the swarm, with the goal of finding the optimal solution. As shown in Fig. 6, the algorithm converges to the optimal solution after 60 iterations.

The experiment formulated above has been repeated for several values of the standard deviation of noise being added to the output of the data in D_t, σ=0,0.1,0.2,…,1. Intuitively, the performance of the model with regard to Q and V degrades with the higher values of σ as illustrated in Fig. 7.

5.1. B. Machine learning data

Here, we report results obtained for experiments completed on publicly available data sets (https://www.mathworks.com/help /matlab/import_export/matlab-example-data-sets.html). The split of the data will be 15 % (D_s), 60 % (D_t: training), and 25 %(D_t: testing). The model formed on the basis of D_s resulted in poor performance with respect to D_t (refer to Fig. 8) once the model was directly transferred.

However, admitting the granulation of models' parameters, the performance index V_{opt} shows improvement on D_t, as shown in Fig. 9.

It is clearly shown that the non-uniform allocation of information granules performs better than the uniform allocation. Among the three protocols, Protocol 3 has the best performance index with respect to V for most datasets, followed by Protocol 2. The average improvement for Protocol 3 over the other protocols is 118.6 % for the uniform allocation of information granularity and 66.5 % for the non-uniform allocation. It is worth noting that relatively small values of r_t can be obtained at $\sigma < 0.2$, leading to good performance of the granular model constructed based on the three protocols. When noise with larger values ($\sigma > 0.2$) is added, the performance of granular models will decrease as s increases, leading to a decrease in the performance index V, bearing in mind that r_t represents a ratio obtained from both domains D_s and D_t.

In comparison, each transfer learning paradigm, including our proposed granular transfer learning approach, has its own set of advantages and disadvantages. By understanding these trade-offs, we can better appreciate the specific strengths of our approach in handling and enhancing adaptability and improving interpretability. Granular models provide a higher level of interpretability compared to traditional numeric models, making it easier for domain experts to understand and validate the results. By representing information granules with intervals and fuzzy sets, our method captures a broader range of potential values, thus enhancing the model's ability to make accurate predictions even in the presence of noisy or incomplete data. Also, the use of interval-valued parameters and allocation of information granularity allows our method to better handle uncertainties and variations in the target domain, leading to improved prediction accuracy.

6. Conclusions

In this study, we have introduced an original concept of granular knowledge transfer. The proposed approach demonstrates that given the obvious fact that D_s and D_t are different, the quality of transferred knowledge is assessed in the form of information granules where the level of information granularity helps quantify the suitability of the originally acquired knowledge (model). The design of the granular model is guided by the optimization of the level of information granularity so that the granular results cover the data in D_s to the greatest extent and produce highly specific results. The experimental results show the performance of the approach through a series of experiments completed for the publicly available Machine Learning data. Of particular interest is the relationship of decreasing values of V when D_t moves away from D_s (the phenomenon is modeled by impacting the data in D_t by the noise of varying variance). It is also worth noting that granular transfer learning is model-agnostic, and as such, it could be applied to any architecture of the model, say a rule-based architecture. For illustrative purposes, the key idea was outlined for interval-valued information granules; the intent was to avoid detailed computing that is more demanding in the case of fuzzy sets or probabilistic granules.

Owing to the generality of the investigations, the detailed pursuits particular to the formal granular environment could be easily carried out. Another augmentation of the study worth pursuing could pertain to the active way of knowledge transfer. Here, we considered a passive way of transfer in the sense that the model in Ds has not been changed but is generalized to its granular counterpart. This way could be expanded by designing a model in D_t with the guidance offered by the granular model developed here in the passive manner, which contributes to the optimization process in the form of a granular regularization component present in the optimized loss function.

Research involving human participants and/or animals

This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent

Informed consent was obtained from all individual participants included in the study.

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CRediT authorship contribution statement

Ahmed Ammari: Writing – review & editing, Visualization, Project administration. **Rami Al-Hmouz:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Medhat Awadallah:** Writing – review & editing, Resources, Project administration. **Witold Pedrycz:** Writing – original draft, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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